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SH.P.K

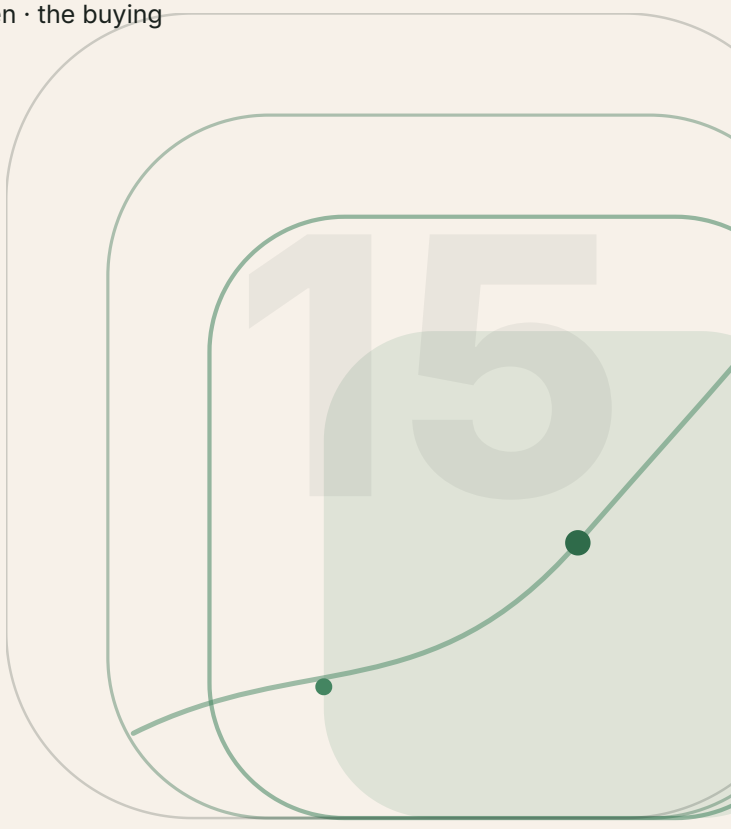
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HARDWARE SHEET HW15 · QUALIFIED BY SPECIFICATION

24 V power supply

The one clean place the modules, the transmitters' loops and the converter take their power from.

QUALIFIES	Qualified by specification. Any 24 V DC supply for a DIN rail, of 30 to 60 W.
USED IN	Kits H, S, F and W (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager who buys it, and the electrician who connects its mains side
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



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HW15

24 V power supply

The 24 V supply is the one clean place the kit's low-voltage parts take their power from: the converter, the two modules and the transmitters' loops, and the box if the box takes 9 to 36 V. It sits on the DIN rail in the enclosure, on the mains side, behind its own fuse or small breaker. Its mains side is the electrician's work. Its 24 V side is the project manager's.

At a glance

WHAT IT DOES	Powers the two modules, the loop transmitters and the converter from one place.
HOW IT QUALIFIES	**Qualified by specification.** Any 24 V DC supply for a DIN rail, of 30 to 60 W.
SPECIFICATION	30 to 60 W
FITTED BY	The electrician connects its mains side. The project manager connects the 24 V side to the converter and the modules.
PLANNING ESTIMATE (EUR)	25 to 50. Not a quote.
USED IN	Kits H, S, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Any make qualifies that meets every line below.

It qualifies if:

- ☐ It gives **24 V DC,** fits a DIN rail, and is rated **30 to 60 W.**
- ☐ The mains voltage on its label is the site's.
- ☐ **Its power covers the kit.** Add up what the labels of everything it feeds say, and take a supply rated for that and half as much again. As a guide: 30 W is enough for the converter, the modules and the loops; take 60 W where the box or the 4G router shares the supply.
- ☐ Its data sheet says that its output is protected against a short circuit and an overload.
- ☐ It is a known make, with its safety approvals on the label, and the electrician who connects it accepts it. It is the part of the kit that is connected to the mains.

STOP

Do not use a plug-in charger or an open supply with bare terminals in the enclosure. **The project manager never connects the mains side:** that is the electrician's work, and it has its own fuse or small breaker.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	One, in the enclosure. It feeds the two modules, the transmitters' loops and the converter, and the box if the box takes 9 to 36 V.
S Solar plant with grid connection	IN THE KIT	1	One. With no module and no loop, it feeds the converter, and the box if the box takes 9 to 36 V.
F Farm or greenhouse with irrigation	IN THE KIT	1	One, as at a hotel: the modules, the loops and the converter.
W Water tank and pump station	IN THE KIT	1	One, as at a hotel: the modules, the loops and the converter.

What it feeds

It feeds	The part takes	When
The converter	9 to 36 V DC	Always
The 4 to 20 mA module	7 to 36 V DC	Kits H, F and W
The pump contact module	The voltage on its label	Where there is a pump to watch
Each transmitter's loop	The loop's current, up to 20 mA	A two-wire transmitter takes its power from the loop
The box	9 to 36 V DC	Only a box that takes 9 to 36 V. A box that takes 12 V keeps its own supply, or a small adapter from 24 V to 12 V
The 4G router, a soil probe, the weather sensor	The voltage on its label	Only if its label allows 24 V

Read every label before a part is connected to 24 V. A part made for 5 V or 12 V is destroyed by 24 V.

Settings, and where they are made

Setting	Value	Where
Output voltage	24 V. Many supplies have a small screw that trims it: leave it at 24 V	Measured with the multimeter, at the office

Fitting

ELECTRICIAN connects its mains side. **PROJECT MANAGER** connects the 24 V side.

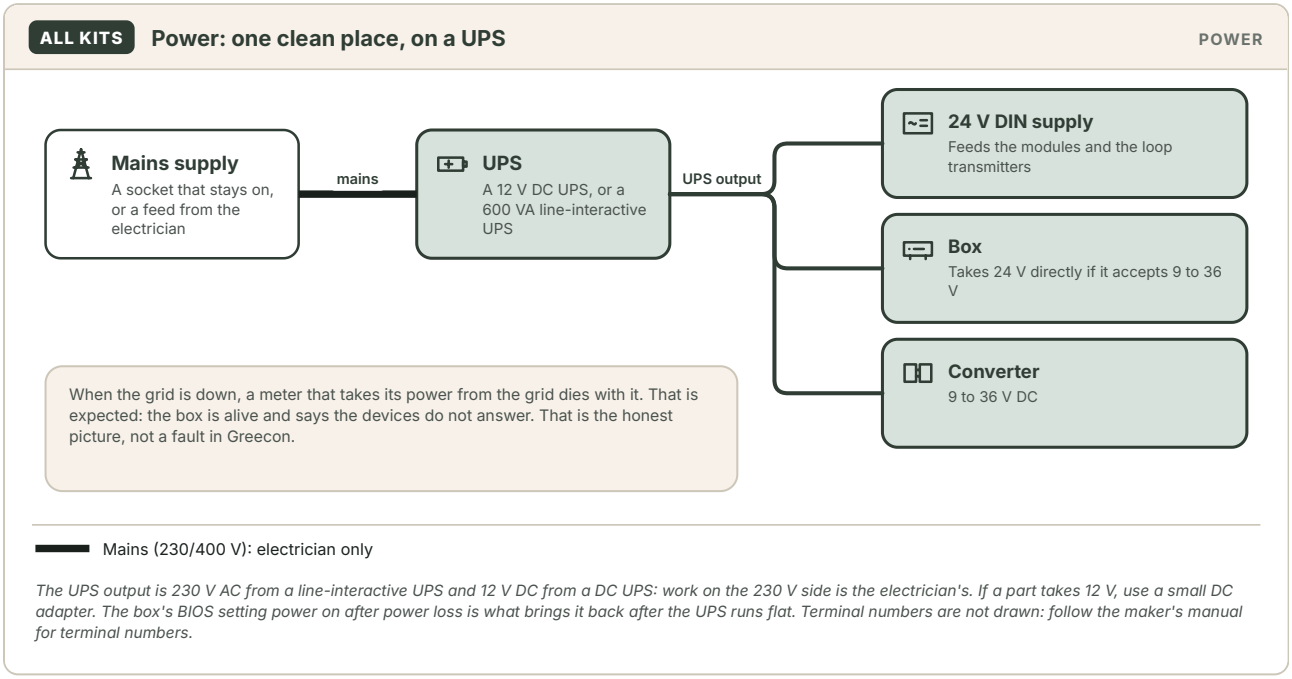


Figure 4-4. Power: everything from one clean place, on a UPS.

- ☐ It sits on the DIN rail, on the mains side of the enclosure, with the air around it that its manual asks for.
- ☐ **The mains comes in through its own fuse or small breaker.** The electrician connects it.
- ☐ Where the site has a mains UPS, the supply is behind it, so that the modules and the loops keep working in a short power cut (sheet HW16).
- ☐ On the 24 V side, plus and minus go to the terminal blocks, and from there to each part. Plus and minus are checked against each part’s label before the supply is switched on.
- ☐ Each transmitter’s loop has the supply in series with the module’s input, as the module’s manual draws it (sheet HW08).
- ☐ Number every wire at both ends.

In the platform

The supply is not in the platform, and Greecon does not read it. What shows that it is working is everything it feeds: the modules answer, and the transmitters read 4 mA or more.

Acceptance test

At the office, on the desk. The electrician makes up the office’s bench lead: the project manager never wires mains.

- [ ] With nothing connected, the multimeter reads 24 V at the output, within a volt.
- [ ] With the whole kit connected and running for an hour, it still reads 24 V, and the supply is not hot.
- [ ] The box finds every device (`greecon-edge find --host 192.168.1.50`).

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
The 24 V side. With everything connected, measure at the supply's output and at the part farthest from it.	24 V at both, within a volt			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
Everything on the 24 V side is dark	No mains at the supply: its fuse or breaker, or the UPS in front of it	The electrician checks the mains side. You do not open it
The supply's light is on, and one part is dark	A loose 24 V wire, or plus and minus the wrong way round at that part	Switch the supply off, and check the two wires against the part's label
The 24 V sags, or the supply clicks on and off	Too much on the supply, or a short circuit on the 24 V side	Disconnect the parts one at a time to find it. A 30 W supply that also feeds the box or the router is changed for 60 W
Every transmitter reads below zero at once	The loops have lost their 24 V	As the first line. One transmitter alone is a wire: see its own sheet

More is in the runbooks (Project Manager Field Manual, PDF 02): R2 (a device does not answer).

Not yet proven

- What each part really draws has not been measured: the guide above is from labels. Measure it at the rehearsal of the first kit, and write the figure down.
- The prices are planning figures, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	